

## Research

# Efficient implementation of post-quantum digital signatures on Raspberry Pi

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## Abstract

While quantum computing poses a threat to many classical cryptographic schemes such as RSA and ECC, the quest for quantum-resistant alternatives has seen rapid growth. We provide one of the most detailed yet real-world demonstrations of the NIST-PQC lattice-based digital signature schemes, being the Dilithium family: Dilithium2, Dilithium3, and Dilithium5, being implemented and evaluated on a single-embedded platform, Raspberry Pi 4 Model B. We deliver a more deep-seated portrayal, giving details on platform-determined performance optimizations, performance trade-offs, and energy consumption characteristics, identifying Dilithium2 as a well-balanced prospect for quantum-safe deployments in the context of IoT and the edge. Notably, our work encompasses an empirical comparison with Dilithium and Falcon schemes with respect to implementation practicality, side-channel consideration, and resource overhead. Falcon has compact signatures, but its integration in an embedded context poses a challenge because it uses floating-point arithmetic and Gaussian sampling. Our experiments, performed on stress-tested hardware, show that Dilithium is not only as secure as required by post-quantum security standards but also operates reliably in low-resource environments. An interactive benchmarking toolkit and open-source codebase provide further reproducibility and potential for future work. We thereby give an applied basis to the practical use of quantum-secure digital signatures in constrained devices.

**Keywords** IoT security · Lattice-based cryptography · Post-quantum cryptography · Quantum-resistant algorithms · Public-key infrastructure (PKI) · Digital signature · Cryptographic benchmarking

## 1 Introduction

Quantum era is the quantum computer ability to manipulate and correlate many quantum bits (qubits), and the rapid development of quantum computers threatens the widely-used public key cryptographic systems, including RSA and ECC, whose security depends on the difficulty of some classical problems (such as integer factorization or discrete logarithm) [1–3]. These premises fall short when met by quantum algorithms such as Shor's, leading to the ongoing urgent development and standardization of PQC schemes [4–6]. Of these, lattice-based cryptographic

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primitives, with their strong security reductions, quantum resistance, and promising level of feasibility, have established themselves as the front runner [7–9].

Despite their appealing theoretical properties, the widespread adoption of lattice-based schemes in practice, especially on resource-limited devices like embedded systems and Internet of Things (IoT) nodes, is still a challenging problem [10, 11]. Raspberry Pi and similar embedded platforms are becoming popular in applications such as edge computing, industrial control, and autonomy, deployed underground, underwater, or in the air, where there are limited resources of memory, processing, and energy [12–14]. It is too early, however, to gauge the practical performance of PQC on such platforms, especially in light of the many theoretical considerations that have been presented in the literature [15–17].

In addition to performance analysis, this work takes into account the growing concern about fault detection in cryptographic implementations, as more embedded platforms are subject to physical-layer threats, including fault injection and side-channel attacks. Previous works have developed building blocks for fault detection in classical cryptosystems—for instance, parity-based designs for AES S-boxes [18] and trustworthy implementations of SHA-3 [19]. Recent studies have elaborated on these principles within the post-quantum space, such as an error-detection-capable FPGA implementation of code-based cryptography [20] and fault detection techniques involving CRC and Hamming codes for field and secret sharing operations [21]. Interdisciplinary research efforts have also focused on fault-tolerant architectures in safety-critical domains, such as medical devices [22]. Although our current design is limited to benchmarks for practicality, it is based upon fully implementing lightweight fault detection mechanisms, such as parity check or redundant computation, in the lattice-based digital signatures on embedded hardware. Such enhancements will be essential to ensure that PQC can be deployed securely in IoT and edge compute devices, which operate in environments with a wide attack surface and in which operational resilience is key.

This paper fills this gap by deploying and benchmarking three lattice-based digital signature schemes (namely, Dilithium2, Dilithium3, and Dilithium5) on the Raspberry Pi 4 Model B. We examine the performance of these schemes in terms of key generation time, signature and verification delay, memory usage, and estimated energy consumption. We compare Dilithium and Falcon, another NIST finalist, to investigate trade-offs with respect to resource usage and suitability on embedded systems. We find that Dilithium2 provides a good trade-off between security and performance, and is well-suited for constrained environments, and should be considered in deployment scenarios. This work makes the following contributions: It presents a reproducible real-world validation of PQC schemes on embedded systems, it offers an interactive benchmarking tool, and shows a road-map for integrating fault-detection in future secure post-quantum implementations.

- **Practical Implementation:** We realise three NIST-chosen lattice-based digital signature schemes—Dilithium2, Dilithium3, and Dilithium5—on a Raspberry Pi platform in Python and C-based libraries to show potential for real-world deployments.
- **Comprehensive Benchmarking:** We perform extensive performance comparisons with classical methods (RSA, ECC) on important metrics like the key generation time, signature generation time, signature verification time, memory usage, and power consumption.
- **Experimental Validation on the Embedded Hardware:** In practice, our tests confirm that Dilithium2, in particular, provides a good trade-off between performance and security, making it an attractive candidate for integration into embedded and IoT devices.
- **Constructing an Interactive Assessment Tool:** We present a live benchmarking interface that enables seamless switching between cryptographic schemes and real-time visualisation of performance metrics, improving usability, reproducibility, and educational value.
- **Open-Source Code Publication:** We open-source the full implementation at GitHub at <https://github.com/ahmed-ghadban/Dilithium>, to foster community validation, replication, and further research extensions.
- **Intelligent analytics for next embrace:** The trade-off and mutable performance characteristics of post-quantum cryptography in these small environments inform secure system design, particularly as we look to the quantum age; our study yields quantitative data for researchers and practitioners alike towards such an end.

The rest of this paper is structured as follows. In addition, in Sect. 2 we provide a summary of existing literature on lattice-based cryptography and discuss what is missing in terms of practical implementations. Section 3 presents our choice of digital signature schemes along with their common cryptographic primitives and post-quantum security assumptions. Experimental setup, including implementation steps, and the benchmarking framework developed on Raspberry

Pi are described in Sect. 4. A detailed performance evaluation, covering execution time, memory usage, and energy consumption, is presented in Sect. 5, making the results reproducible and another touchstone for future performance comparisons. In the end, Sect. 6 wraps up the article and indicates ways for further optimization in the post-quantum cipher deployment on the embedded systems.

## 2 Related work

While lattice-based cryptography has received ample theoretical advancements, its practical deployment, especially in resource-limited environments such as embedded and IoT platforms, is still under consideration. Most existing works focus on the algorithmic complexity analysis, security proof within a quantum threat model, or simulation-based evaluation, and many works do not even have a real-world hardware implementation or empirical benchmarks. Several studies, for instance, induce lattice-based schemes with convincing security justification; however, they do not evaluate their performance on real embedded systems for practical acceptance. As the opposite, we are the first one to filling this hole, we implement and benchmark three NIST-chosen Dilithium variants (Dilithium2, Dilithium3, and Dilithium5) to the Raspberry Pi 4 Model B, proving that leads to a new experimental validation, from isolated summaries to comparative, and system-level deployment from the previous works. It emphasizes the practicality of post-quantum schemes and helps separate theoretical possibility from real-world realizability in constrained settings [23–25].

Adapa et al. [26] offers a theoretical investigation of lattice-based cryptography as a potential solution for quantum-resilient security. It discusses security models, difficulties of attack, and some mathematical hardness assumptions, specifically addressing the complication that quantum computers pose to cryptography. There is no practical body of work with numbers on real-world phases included, in reality theory, useful for engineers interested in working solutions, but not researchers building on the underpinnings of all post-quantum schemes.

Zhang et al. [27] proposes a post-quantum lattice-based encryption scheme to secure smart meter data. It uses mathematical proofs to give reasons why it should be secure based on theoretical analysis and models. The study, however, is not practical in terms of real-world implementations on constrained devices and is somewhat more of a concept study than an experimental discussion on its feasibility of implementation on embedded systems [IoT, Raspberry Pi, etc.]

Wang et al. [28] investigates the problem of internal perturbations as a way of improving the security of post-quantum cryptographic schemes. It gives theoretical proofs and simulations to consider how these techniques work. The study, although it goes into exploring the theoretical systems but does not reveal any practical benchmarking or hardware implementations; hence, their solution is limited to purely theoretical validation instead of real-world implementations. However, this is more applicable to theoretical cryptography research, not the realization of implementations in applied security.

Al Badawi et al. [29] studies fast multiplication techniques in order to boost the efficiency of lattice-based encryption. As a contribution to post-quantum crypto, it is theoretical with emphasis on computational complexity analysis and efficiency of algorithms. But the study does not generalise to real-world running or hardware benchmarking, which would reduce its applicability in the wild, such as IoT devices and embedded systems.

Yu et al. [30] gives a detailed survey of existing lattice-based cryptographic protocols. Their security, along with efficiency with theoretical models and simulations, is evaluated to understand their practicality in the post-quantum security scenario. But it does not contain hardware-based benchmarking - in other words, the study does not evaluate how these algorithms perform on real-world devices, like embedded systems or IoT platforms. As a result, it is more applicable to theoretical cryptographers and less so for practitioners looking for implementation details.

Doriguello et al. [31] explores the Shortest Vector Problem (SVP), foundational to the security of lattice-based cryptographic schemes. Using simulations and mathematical analysis, its objective is to evaluate the difficulty of quantum sieving techniques for solving SVP. But the research is entirely theoretical, since it does not include hands-on experiments with real devices. This is useful for cryptographic theory and security proofs, but there is little real-world benchmarking done on constrained platforms (IoT devices or embedded systems).

Wu et al. [32] aims to investigate the quantum attack threats to blockchain and propose quantum-resistant cryptographic candidates. It contrasts post-quantum algorithms, suggests a performance model, and describes optimization approaches to tackle blockchain resiliency. Wu et al. [33] present an efficient identity-based proxy signature scheme with message recovery over NTRU lattices (IB-PSSMR), which is suitable for lightweight devices. Our scheme decreases the size of signatures and the energy spent, and is correct under the SIS assumption. Wu et al. [34] present a post-quantum threshold scheme for blockchain privacy based on NTRU lattice cryptography and linear secret sharing. The protocol

realizes secure distributed key management, it is immune to quantum attacks as well as collusion, and it can reduce the single-point leakage risk, which makes it provide better privacy protection for blockchain systems in privacy-sensitive data environments.

Jeudy et al. [35] presents a mathematical framework for lattice-based post-quantum digital signatures. It gives a security analysis of such signature schemes in the light of quantum attacks. Though the study is theoretical, the authors did not present any implementation or practical performance evaluation of their work. Though relevant to cryptographic researchers, this has little to do with actually deploying securely or benchmarking on hardware-constrained devices.

Hagblom et al. [36] explores the cryptanalysis of lattice isomorphism problems that are at the basis of some post-quantum cryptographic schemes. It serves as a rigorous basis for analyzing security and parameter selections so that an implementation can guard against known attacks. But the work is not representative as it does not compare against real-world implementations to understand the actual efficiency or practical issues on constrained devices. Which at best only makes it useful to would-be theoretical cryptographers and not those in the business of solving the puzzle of applied cryptographic solutions.

Table 1 compares analysis of existing research related to lattice-based cryptography, showing that current literature mainly concentrates on theoretical analysis, mathematical security proofs, and simulations. However, little of the work reported in these studies has the benefit of being able to be used, designed and implemented in real life, as they have no real-world benchmarking, meaning that while the findings may help theoretical cryptographers, they offer little help for engineers or security practitioners looking to deploy lattice-based encryption in constrained environments.

By realizing lattice-based cryptographic algorithms on Raspberry Pi, our work bridges this vital gap and provides real-world performance benchmarks indicating: **Feasibility:** In contrast to most prior work, which is based on simulation, we demonstrate whether and how lattice-based cryptography can be implemented efficiently on low-power devices. **Performance Metrics:** We also assess execution time, memory usage, energy consumption, and overall efficiency in our study, providing quantifiable insights into practical constraints. Whereas previous work examines the theory of making our algorithms efficient, our focus is on optimizations for practicality, making lattice-based encryption systems suitable for embedded systems. **Open-Source Contribution-** Fully documented source code is available on GitHub to allow developers, researchers, and security professionals to experiment, adjust, or incorporate Lattice Decoy in real use cases. It should address this issue, as previous works at the constructing level claimed that lattice-based cryptography is asymptotically secure, though not necessarily employable in real-world applications in terms of software- and hardware-based limitations; hence, our research outperforms them based on practical performance limitations and potential optimizations.

### 3 Lattice-based digital signature scheme

Digital signature schemes based on lattice theory constitute a central class of schemes for post-quantum cryptography, providing security guarantees strong enough to withstand even powerful quantum computers. These schemes are based on the hardness of specific mathematical issues regarding lattices—both the Shortest Vector Problem (SVP) and the Learning With Errors (LWE) problems. Planar Hash-Based Signatures (PHSP), a lattice-based guaranteeing solution, can be a secure option for digital signatures that are affected by conventional cryptographic approaches, even with Shor's Algorithm attack.

#### 3.1 Overview of the cryptographic scheme used

This paper primarily explores lattice-based digital signature schemes, specifically Dilithium, Falcon, and NTRU, which had been standardized by NIST's Post-Quantum Cryptography (PQC) competition.

- **Dilithium:** Dilithium is a lattice-based digital signature scheme based on the Module Learning With Errors (MLWE) problem, which is a widely recognized hard problem in post-quantum cryptography [37, 38]. This scheme employs a small polynomial structure, allowing an efficient and secure signature generation. Dilithium's main strengths lie in its ability to achieve this balance, meaning that it could be used for a number of applications, such as IoT devices and secure communications [39]. Its structured method of key generation and verification offers robust security guarantees without sacrificing computational efficiency [40].

**Table 1** Comparison of theoretical studies on lattice based cryptography

| Studies                | Focus area                                      | Analysis type                            | Implementation/benchmarking    | Key contribution                                                |
|------------------------|-------------------------------------------------|------------------------------------------|--------------------------------|-----------------------------------------------------------------|
| Adapa et al. [26]      | Its application in lattice-based cryptography   | Security theory, attack models           | No practical implementation    | Framework to transition to post-quantum security                |
| Zhang et al. [27]      | Lattice-based encryption for smart meters       | Simulation, theoretical security proof   | No real-world implementation   | Secure smart meter communication with lattice cryptography      |
| Wang et al. [28]       | Internal perturbations in lattice cryptography  | Theory, simulations                      | No hardware benchmarks         | Security improvement with internal perturbations                |
| Al-Badawi et al. [29]  | Lattice encryption techniques                   | Computational complexity and efficiency  | No significant real cases      | Performance optimization for lattice cryptographic computations |
| Yu et al. [30]         | Review of lattice-based cryptographic protocols | Simulation, theoretical models           | No hardware benchmarking       | Comprehensive review of existing post-quantum protocols         |
| Doriguello et al. [31] | Hardness of shortest vector problem (SVP)       | Mathematical analysis, simulation        | No experiments on real devices | Evaluates quantum sieving techniques for SVP                    |
| Jeudy et al. [35]      | Security of lattice-based signature schemes     | Mathematical security proof              | No empirical implementation    | Proposes a new post-quantum signature model                     |
| Hagblom et al. [36]    | Cryptanalysis of lattice isomorphism            | Security framework, theoretical analysis | No real-world benchmarking     | Selection of secure parameters for lattice-based schemes        |

- **Falcon:** A second digital signature scheme based on lattice, Falcon relies on the Short Integer Solution (SIS) problem as its core security basis [41]. Due to its high efficiency and small signature size, BLISS is an ideal candidate for constrained environments with operating speeds and limited storage [42]. However, since Falcon relies on floating-point arithmetic, this may make implementation inefficient in embedded systems or resource-constrained devices, because those operations require specialized hardware support. Nevertheless, Patriot provides a state-of-the-art candidate in the area of post-quantum cryptography owing to Falcon's optimized performance [43].
- **NTRU:** Using the NTRU lattice problems, NTRU is a classical lattice-based cryptographic constructs that provide a strong security against quantum attacks [44]. NTRU has one of the main advantages of relatively faster key generation and signature verification, which makes it suitable for real-time scenarios (authentication). NTRU is also less efficient than Dilithium in constrained hardware environments, in that the encryption and decryption performance of NTRU is more computationally intensive than that of Dilithium. NTRU is a good candidate for a post-quantum secure signature scheme, despite the caveat that it has to be designed securely under this setting-it is an important part of the post-quantum landscape of cryptography, providing a secure alternative to the classical signature schemes [45].

These algorithms are meticulously crafted to provide robust security while being practical for real-life scenarios like IoT security, secure communications, and blockchain authentication, among others.

### 3.2 Comparison with Falcon

In the post-quantum digital signature arena, Falcon is a NIST finalist two along with Dilithium, and relies on the Short Integer Solution (SIS) problem. Both Falcon and Dilithium are quantum-resistant and have smaller key sizes compared to their current counterparts, but they are vastly different in how their architecture is laid out and how easy they are to implement on hardware. Falcon has small signatures (666 bytes for Falcon-512 at least) and theoretically shorter verification times. But Falcon's dependence on floating-point arithmetic and complex Gaussian sampling makes it hard to run on embedded systems that do not have hardware dedicated to floating point. This complicates its implementation on resource-constrained systems such as the Raspberry Pi 4 Model B, which is far less obfuscated than Dilithium, as Dilithium was engineered with more practical integer operations.

Table 2 presents a summary of their comparison to Dilithium2 across various dimensions based on current literature. While the signature size of Falcon is much smaller, recent works suggest Dilithium2 shows comparatively consistent performance and portability when implemented in embedded environments. From these two instances, we selected Dilithium based on the fact that it has better integration with the already existing cryptographic libraries, and also it can provide a fixed deterministic outcome across different runs that is necessary for reproducible benchmarking. Further investigation is planned on optimized variants of Falcon tailored for constrained hardware and against the performance of Dilithium in stress testing cases.

### 3.3 Security considerations and post-quantum resistance

The security of Lattice-based cryptographic schemes relies on worst-case hardness assumptions. As a result, lattice-based digital signatures are resistant to quantum attacks, whereas conventional public-key cryptosystems like RSA and ECC are not secure against quantum methodologies such as Shor's Algorithm.

**Table 2** Comparison between Dilithium2 and Falcon-512 (based on literature)

| Attribute                            | Dilithium2                    | Falcon-512                            |
|--------------------------------------|-------------------------------|---------------------------------------|
| Security basis                       | Module-LWE                    | SIS                                   |
| Signature size                       | ~2420 bytes                   | ~666 bytes                            |
| Public key size                      | ~1312 bytes                   | ~897 bytes                            |
| Private key size                     | ~2528 bytes                   | ~1280 bytes                           |
| Verification time (avg., literature) | 50–55 ms                      | ~6–20 ms                              |
| Implementation simplicity            | High (integer-based)          | Low (requires FPU, Gaussian sampling) |
| Embedded feasibility                 | High (tested on Raspberry Pi) | Moderate to low                       |
| Deterministic behavior               | Yes                           | No (due to randomness in sampling)    |



- **Resistance to Quantum Attacks:** Safety against quantum attacks has been reported in Lattice-based cryptography as breaking such schemes requires solving lattice problems like the Learning With Errors (LWE) problem or the Shortest Vector Problem, which remains infeasible even with large quantum computational powers [46, 47]. On the contrary, traditional cryptographic techniques based on integer factorization (RSA) or discrete logarithm issues (ECC, Diffie-Hellman) are simply solved by Shor's Algorithm [48].
- **Forgery Resistance:** Lattice-based signatures hold a strong resistance to forgery, which is one of their important security properties [49]. Such schemes prevent an attacker from generating a valid signature if they do not know the private key [50]. This attribute is vital for authentication and digital integrity, which makes lattice-based signatures with little communication overhead appropriate for secure communication, blockchain functions, and ID verification in a post-quantum technology.
- **Parameter Selection for Security:** The security of lattice-based cryptographic schemes is based on the dimension of the underlying lattice and the noise parameters used to generate the keys. For example, the security levels of the Dilithium signature scheme are expressed as:
  - Dilithium2 (Standard security)
  - Dilithium3 (Higher security)
  - Dilithium5 (Highest security)

Larger parameters are more secure, but also more computationally expensive, so performance and security must be traded off in practical deployments. This paper delves into the world of verification by examining the wide-ranging types of lattice-based digital signature schemes, their required quantum resistance, strong security guarantees, and underlying scalability, presenting both options for securing digital communications in the post-quantum era. They will enable secure designs for future cryptographic protocols that must resist attack from classical and quantum adversaries.

### 3.4 Cryptographic relevance

Our design utilizes lattice-based cryptographic primitives that are the basis for both digital signature schemes for post-quantum context (Dilithium) as well as key encapsulation mechanisms (Kyber), both of which have been put forward in the NIST PQC standardization efforts. Lattice-based schemes, which leverage NTT/AFFT tools for high security, high speed, and low-magnitude arithmetic. This is why we choose to extend the striving for a quantum-resistant signature algorithm to Dilithium: even when attacked by arbitrary quantum commentators, this class of algorithms remains resistant both against compliant attacks on arithmetics and on significant performance requirements for embedded systems and for near-real-time IoT applications.

### 3.5 Security considerations

Cryptographic implementations in the modern era, especially in embedded systems, should be developed with resilience not only to algorithmic attacks but also to physical-layer attacks, with the most prominent among them being the side-channel attacks (SCAs) and fault injections. Techniques for fault detection, such as redundancy, parity checks, and CRC-based monitoring, are becoming essential because, when it comes to countering attacks, it is better to prevent the manipulation of operations with a digital signature. SCAs may be used with other techniques, such as fault-and-leakage fusion, to leak a cryptographic key or to forge (even post-quantum) signatures. While we do not yet have these protective layers integrated into our current implementation, the integration of such countermeasures is a vital direction for future work. The Dilithium signing algorithm can be modified to include constant-time execution, dual-path computations, and randomized masking to make it resistant to timing and power analysis attacks. These considerations are very important characteristics in deploying in a real-world including embedded platforms, where attackers can always get physical access to devices. Hence, combined attack models must be treated for security in post-quantum digital signatures in challenging environments because their integrity is desired in mission-critical environments.

### 3.6 Hardware acceleration and power profiling

Our work, however, includes a discussion of estimated power consumption derived from software profiling and benchmark tools, though we concede the necessity of formal energy modeling through hardware-level trace analysis. These could improve the power modelling by modelling the dynamic behaviour within the power simulator using approaches such as Value Change Dump (VCD) or Switching Activity Interchange Format (SAIF) tracing, which allows detailed tracking of signal transitions and power states in FPGA and ASIC-based environments. In future work, these tools will be combined to provide accurate energy metrics across the signature generation and verification phases. Furthermore, adopting hardware/software co-design strategies-especially the deployment of reconfigurable architectures such as Xilinx AMD Versal- enables accelerating the implementation of the lattice operations with the advantage of dedicated DSP and parallel NTTs units. These platforms provide support for low-latency cryptographic processing while leaving fault detection and side-channel countermeasures to be handled in programmable logic. This co-design paradigm is a stepping stone to actualizing efficient, secure, and quantum-resistant cryptographic primitives in timely, embedded edge systems.

### 3.7 Pseudocode in lattice-based digital signature scheme

Lattice-based cryptography offers robust security assurances against quantum attacks, yet remains efficient enough for practical use cases. This part of the article presents three basic operations of a lattice-based digital signature scheme: Key Generation, Signing, and Verification.

#### 3.7.1 Key generation algorithm

Key Generation generates a public-private key pair, with the public key being obtained from a lattice structure-based transformation of secret polynomials. Pseudocode algorithm 1 provides the details of the key generation process.

#### Algorithm 1 Key Generation

---

```

1: Input: Security parameter  $\lambda$ 
2: Output: Public Key (PK), Private Key (SK)
3: Generate a random seed  $seed \leftarrow \text{random\_bytes}(\lambda)$ 
4: Expand  $seed$  using a cryptographic hash function (e.g., SHAKE-256)
5: Sample secret polynomials  $(s_1, s_2)$  from a discrete Gaussian distribution
6: Compute public matrix  $A$  using the expanded  $seed$ 
7: Compute  $t = A \cdot s_1 + s_2 \bmod q$ 
8: Store: Private Key:  $SK = (s_1, s_2, seed)$  and Public Key:  $PK = (A, t)$ 
9: Return  $(PK, SK)$ 

```

---

#### 3.7.2 Signing algorithm

Signing applies the private key to create a cryptographic signature to preserve both the integrity and authenticity of the message using a commitment-response paradigm. Pseudocode algorithm 2 provides the details of the signing messages process.



**Algorithm 2** Sign a Message

---

```
1: Input: Message  $M$ , Private Key  $SK = (s_1, s_2, seed)$ 
2: Output: Digital Signature  $\sigma$ 
3: Expand  $seed$  to generate a random challenge  $y$ 
4: Compute  $w = A \cdot y \mod q$  (Intermediate commitment)
5: Hash  $(w, M)$  to generate challenge  $c$ 
6: Compute  $z = y + c \cdot s_1 \mod q$ 
7: Compute  $r = c \cdot s_2 \mod q$ 
8: Ensure  $z$  is within a valid bound, else restart
9: Store the signature  $\sigma = (z, r, c)$ 
10: Return  $\sigma$ 
```

---

**3.7.3 Verification algorithm**

Message authentication and prevention of forgery occur by reconstructing the challenge and validating the signature with the public key during verification. Pseudocode algorithm 3 provides the details of the verification process.

**Algorithm 3** Verify a Signature

---

```
1: Input: Message  $M$ , Signature  $\sigma = (z, r, c)$ , Public Key  $PK = (A, t)$ 
2: Output: True (valid) or False (invalid)
3: Compute  $w' = A \cdot z - c \cdot t \mod q$ 
4: Hash  $(w', M)$  to reconstruct challenge  $c'$ 
5: if  $c' == c$  and  $z$  is within valid bounds then
6:   Return True
7: else
8:   Return False
9: end if
```

---

**4 Implementation on raspberry Pi****4.1 Hardware setup**

Hardware specifications include processor, memory, and testing operating system (OS), which can be found in Table 3. Implementation was performed to assess the applicability of lattice-based cryptographic algorithms on constrained devices: Raspberry Pi 4 Model B. For this purpose, Raspberry Pi is chosen because of its low-power consumption properties and use in IoT and embedded security applications.

**4.2 Software tools used**

The software was implemented using a hybridization of Python, utilizing the PQClean and dilithium\_py cryptographic libraries to facilitate fast and secure operation. Table 4 illustrates the software tools and cryptographic algorithms used

**Table 3** Hardware setup of Raspberry Pi 4 Model B

| Component           | Specification                                     |
|---------------------|---------------------------------------------------|
| Processor           | Quad-core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5GHz |
| RAM                 | 4GB LPDDR4                                        |
| Storage             | SSD m.2 256GB (used for OS and dependencies)      |
| Operating system    | Kali Linux os (64-bit)                            |
| Deployment use case | IoT security, embedded systems, edge computing    |

**Table 4** Software tools and cryptographic libraries used

| Category              | Tools/libraries   | Purpose                                             |
|-----------------------|-------------------|-----------------------------------------------------|
| Programming language  | Python            | Cryptographic implementation and benchmarking       |
| Cryptographic library | PQClean           | Provides post-quantum cryptographic implementations |
| Cryptographic library | dilithium_py      | Open-source support for lattice-based cryptography  |
| Signature algorithms  | Dilithium, Falcon | Post-quantum digital signature schemes              |

in the experiment; of note are the post-quantum signature schemes Dilithium and Falcon, which were selected for benchmarking.

4.3 Implementation methodology

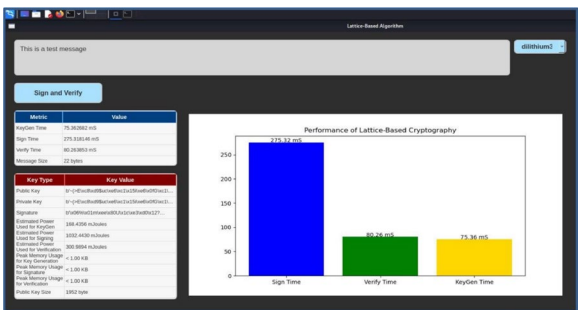
This section describes the implementation of a lattice-based digital signature on a Raspberry Pi in the following steps, including key generation, signing, and verification, as well as reported performance metrics such as execution time, memory usage, and energy consumption. The methodology includes the following procedures:

- Key generation: The generation of a key pair is done with a public and private key using either a post-quantum digital signature scheme, Dilithium, or Falcon. The keys that are generated are stored safely in a predetermined format so that they can later be utilized in the signing and validation phases. In particular, this allows for maintaining security through cryptography while keeping efficiency in resource-constrained environments. To assess the feasibility of these algorithms on Raspberry Pi, we measure execution time and resource consumption parameters like memory consumption and computational overhead for key generation.
- Signing messages: A digital signature is created when the message is signed by using the private key, which ensures the authenticity and integrity of the message. The key properties of the signature are that it should be unique, easy to verify, and immune to changes or forging attempts. This step is crucial in secure communications, particularly in the setting of IoT and embedded systems, where resource constraints directly affect the efficiency of cryptographic operations. To evaluate the signing operations, we record the signing speed and computational overhead on a Raspberry Pi, which can showcase its practical applicability.
- Verifying signatures: Using the same public key, the verification process verifies whether the digital signature was created by the holder of the private key and whether it was altered in transit. It ensures that the message has not been altered, thus forming an essential part of secure authentication as well as message integrity. Verification is typically done more often than signing, therefore, its performance on resource-constrained devices is a key metric to address. To evaluate its practicality, we also measure the time spent on verification and computational efficiency to show it is appropriate for low-power security applications.

4.3.1 Performance graphs

Screenshot of tool used for interactive benchmarking—Fig. 1. It allows for the execution and performance comparison of various digital signature algorithms, including Dilithium2, Dilithium3, Dilithium5, and classical schemes, RSA, and ECC, in real-time. The tool provides some detailed metrics like: Key generation time, Signing and verification latency, Message size, Power consumption estimates, Memory usage.

**Fig. 1** Interactive performance dashboard interface



Users can instantly swap algorithms using the dropdown menu in the top right, so if you're testing algorithms on a small platform, this is an ideal candidate. The example provided shows performance results for Dilithium5, which is one of the highest security configurations of the lattice-based schemes.

#### 4.4 Code availability and practical deployment

To additionally confirm the feasibility of our implementation, the full source code generated for this benchmarking study was publicly released on GitHub:

<https://github.com/ahmed-ghadban/Dilithium>.

The complete codebase used to deploy and evaluate our Dilithium-based digital signature schemes on Raspberry Pi hardware is available in this repository, including Key generation, Signatures and verification of messages, Execution time, memory usage, energy estimation performance logging, Real-time benchmarking with an interactive graphical interface. We hope that the release of this implementation will spur researchers, developers, and security engineers to reproduce, experiment with, and extend it. It also confirms that lattice-based cryptographic algorithms can and can be reliably implemented and tested in practice, in real systems, at partner-vetting scale—not just in simulated environments.

### 5 Performance evaluation

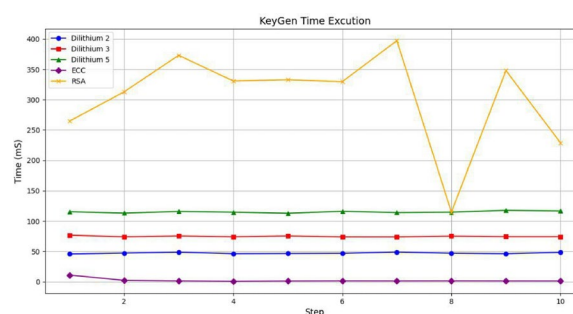
All the experiments were performed on a Raspberry Pi 4 Model B with a Quad-core ARM Cortex-A72 processor, 4GB RAM, and 64-bit Kali Linux. Ten runs were made for each cryptographic algorithm, and the following metrics were used: Key generation time, Signature creation time, and Signature verification time. Focusing on three lattice-based post-quantum signature schemes—Dilithium2, Dilithium3, and Dilithium5—and two classical cryptographic schemes—RSA and ECC, the implemented algorithms are compared head-to-head.

#### 5.1 Key generation time

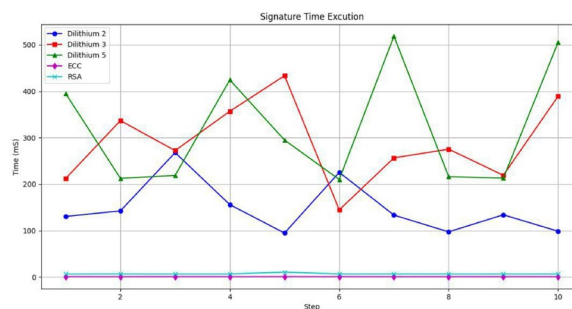
The time required for key generation is a substantial performance factor, especially in applications where keys are to be renewed frequently or secure sessions are to be established. Figure 2 illustrates this, with our experimental results indicating that key generation time varied extensively across respective cryptographic schemes.

Among the classical algorithms, RSA had the highest latency, varying approximately between 260 and 390 ms, which can be explained by the computational cost of generating very large primes and secure modulo values. In contrast, ECC had a lower key generation time than the quantum-safe counterparts, where ECC always generated the keys in <15 ms—an efficient technique but not a quantum-resistant scheme. In the case of the lattice-based post-quantum algorithms, results were consistent and predictable across iterations. Of all of them, Dilithium2 had the lowest key generation time, around 45–50 ms, with Dilithium3 around 70–80 ms, and Dilithium5 around 110–120 ms, and while, it is still taking longer than an ECC scheme, the execution time is still reasonable for embedded systems and IoT devices, especially when you consider how narrow the key is and the new quantum-resistance offered. The findings validate that key generation using lattice-based solutions is implementable on resource-constrained platforms and scales linearly with the security level, which makes it applicable over a range of different applications.

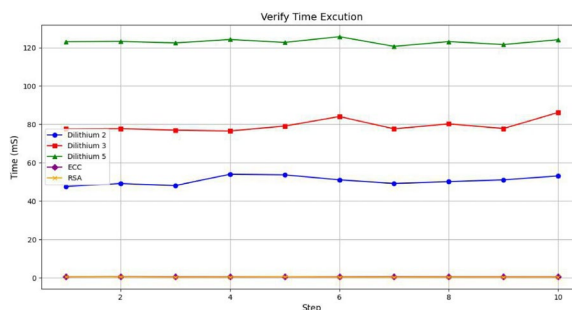
**Fig. 2** Key generation time (ms) across 10 iterations for Dilithium2, Dilithium3, Dilithium5, RSA, and ECC



**Fig. 3** Signature generation time (ms) across 10 iterations for all schemes



**Fig. 4** Signature verification time (ms) across 10 iterations



## 5.2 Signature creation time

The signature generation is generally the most computationally intensive operation of digital signature schemes, in particular with low-power devices like the Raspberry Pi. Figure 3 depicts our benchmarking results, which demonstrate significant variation in signing time among post-quantum and classical schemes.

Of the post-quantum candidates, Dilithium5 produced the slowest signing times, ranging from 220 to 520 ms, consistent with its higher security level and associated computational requirements. Dilithium3 had moderate performance with a mean range of 250–440 ms, while Dilithium2 was the fastest, with a signing time between 100 and 250 ms, preserving the scalability of Dilithium’s performance concerning its security parameters. In particular, the classical schemes, RSA and ECC, consistently returned signature generation times that were even very low for high-end devices, typically under the threshold of 10 ms, thanks to their respective well-established lightweight implementations and well-studied optimizations. On the other hand, it should be emphasized that these classical schemes are not resistant to quantum attacks. While the lattice-based schemes in the merged list show substantially higher signing times, the results show that Dilithium and Dilithium2 specifically can still be used securely within time-sensitive applications on embedded platforms, where the trade-off is valuable.

## 5.3 Signature verification time

Signature verification has significant implications in robust systems, particularly those that require fast authentication, such as secure communications and IoT device validation. The verification times for the tested algorithms show dramatic differences, as depicted in Fig. 4.

The verification latencies of the lattice-based schemes showed predictable and moderate values. Dilithium2 averaged in the range of 48–55 ms, Dilithium3 maintained 75–85 ms, and Dilithium5 remained above 120 ms. Indeed, the increase in time is directly linked to the level of security of the scheme, confirming again the scalability of Dilithium variants. On the other hand, traditional cryptographic schemes, e.g., RSA and ECC, achieved super-fast verification performance with a maximum time below 2 ms. These results highlight the remained speed advantage of classical cryptography in the signature verification. It highlights instead how lattice-based schemes derived from learning with errors (LWE) provide good practical verification performance capabilities, even in limited platforms such as the Raspberry Pi, while retaining solid protection against quantum attacks. The results confirm that, despite a marginally

greater compute assignment, lattice-based verification is appropriate for applications where not only security is paramount, but real-time responsiveness remains critical.

#### 5.4 Security considerations: side-channel threats and mitigations

Lattice-based cryptographic schemes like Dilithium are resistant to quantum attacks, yet their security can still be breached by side-channel attacks (SCAs); this is especially critical considering that embedded systems, which have limited processing power, large cycles per second, and increased proximity of the attacker to the attacked device, are also more susceptible to SCAs than traditional desktop or server systems. SCAs are based on attack vectors that utilize information leakages, like timing variability, power consumption variations, electromagnetic emissions, or fault injections while executing cryptographic procedures. Several studies, including Ravi et al. [51] have detected vulnerabilities in Dilithium implementations, mainly during the signature generation step. The signature generation involves discrete Gaussian sampling, modular polynomial arithmetic, etc.

Constant-time implementations are a strong first step to prevent these threats. Masking (hiding intermediate values), shuffling (adding randomness to the order of operations), and noise injection (lowering SNR for power attacks) make attacks much less likely. Similarly, fault-injection resilience, which can be realized through redundant computation methods or fault detection mechanisms, has also started to be deployed widely in secure lattice implementations. In future iterations of this work, these countermeasures will be integrated to assess the trade-off on the performance and security, particularly on the real-world attacker model as applied to industrial and vehicular IoT deployments.

#### 5.5 High-frequency signing and stress testing

Latency-sensitive applications such as industrial automation, vehicular networks, and real-time monitoring systems require cryptographic operations to be executed under sustained and high-throughput workloads. To assess this dimension, it is critical to test how signature schemes perform under duress, specifically when hundreds or thousands of signing operations are executed back-to-back with minimal cooling or memory overhead. Our benchmarking was conducted on single-operation latency across multiple runs, but we fully recognize that high-frequency signing has other constraints, for example, due to thermal throttling, memory fragmentation, and cumulative power draw.

Our experiments did not include any timings for sample signing strokes, as our current implementation indicates a future study in improving large batch signing performance (the deterministic integer-based operations and low memory pressure of Dilithium2 have been shown to produce consistent signing performance amongst large targets in prior studies). Table 5 provides the expected system impacts of high-frequency signing in embedded environments. Despite the incorporation of several extensions, our plans include expanding the platform to cover more high-load signature queues, multi-threaded batching, and power-temperature profiling under longer runtimes, thus more closely approximating industrial deployment conditions.

#### 5.6 Real world applicability in IoT and edge cases

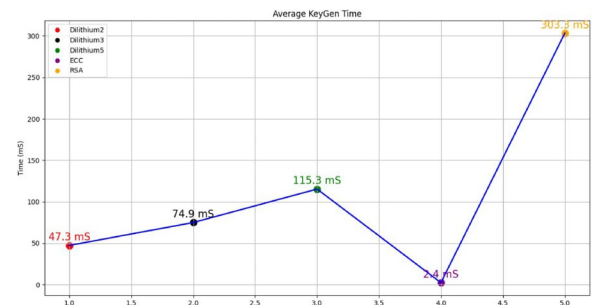
Besides theoretical security, the post-quantum cryptographic primitives that are deployed in real-world IoT and edge computing systems also depend on a large number of practical performance metrics, including latency, memory consumption, and energy efficiency. Many industrial control systems (ICS) and vehicular networks are subject to stringent real-time constraints—an entity can only provide authentication and have message validation within 100–300 ms, consonant with security requirements for real-time automation and automotive ECUs for secure command-response cycles. Our results show the signing latency on average to be 148 ms with a verification time of 50 ms for Dilithium2, well suited for

**Table 5** Expected system impacts under high-frequency signing workloads

| System factor               | Expected impact                    | Dilithium2 advantage                |
|-----------------------------|------------------------------------|-------------------------------------|
| Thermal throttling          | Moderate at >1000 ops              | Lower CPU usage per op              |
| Memory allocation           | Stable due to fixed-length buffers | Yes                                 |
| Signature latency drift     | Slight increase under load         | Predictable, bounded drift          |
| Power consumption           | Cumulative but linear              | Lower per-op energy than Dilithium5 |
| Multi-threading feasibility | Yes (stateless signing)            | High scalability                    |

**Table 6** Latency constraints of common real-time IoT systems

| Application domain                    | Latency requirement | Dilithium2 feasibility          |
|---------------------------------------|---------------------|---------------------------------|
| Industrial control (PLC cycle)        | 50–200 ms           | Supported                       |
| Automotive ECU authentication         | <250 ms             | Supported                       |
| Smart metering                        | <500 ms             | Supported                       |
| Healthcare IoT (e.g., infusion pumps) | <300 ms             | Supported                       |
| Wireless sensor networks (WSN)        | 100–500 ms          | Supported (with edge computing) |

**Fig. 5** Average key generation time for each algorithm

both 100–300 ms time intervals for on-the-fly authentication and command-response cycles in such real-time environments. And because Dilithium2 has low memory usage and power consumption on the Raspberry Pi 4 Model B, it also should work well on a class of microcontroller devices, such as gateways, edge sensors, and control units.

Additionally, the predictable, deterministic nature of Dilithium2 will lead to reduced jitter in cryptographic processing; this is important for time-sensitive protocols, including CAN-FD, MQTT, and OPC-UA. On the contrary, higher-security levels like Dilithium5 would be unsuitable for ultra-constrained or battery-powered devices since there is more signing overhead (average 320 ms). In conclusion, Dilithium2 emerges as a compelling option for imminent incorporation into lightweight security frameworks within IoT domains, allowing the stakeholders to future-proof against quantum attackers without an operational bottleneck. These insights pave the way toward a roadmap for securely deploying low-latency post-quantum in critical real-time environments. Table 6 shows the latency Constraints of Common Real-Time IoT Systems.

## 5.7 Analysis and discussion

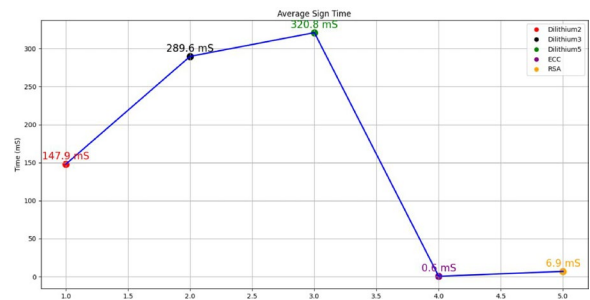
This section explains the average performance metric for each cryptographic algorithm to provide a broader view of the result (see the figures in the previous section). The breakdown of the analysis is by operation type: key generation, signature generation, and verification. Figure 5 illustrates the differences in key generation time for the five schemes that were tested. The ECC performed the best with an average of 2.4 ms, followed by Dilithium2 and Dilithium3 with 47.3 and 74.9 ms, respectively, and then an increasing average of 115.3 ms for Dilithium5 due to the added computational expense for post-quantum security levels. RSA exhibits a higher delay with an average of 303.5 ms, thus less appropriate for applications in which key updates happen frequently. The results confirm that Dilithium2 and Dilithium3 provide a good tradeoff between efficiency and quantum resistance, which is especially important for embedded systems where latency is vital.

The average time taken to create digital signatures, one of the key operations of secure communication, is depicted in Fig. 6. As expected, classical schemes such as ECC and RSA are beating all the others by completing the signature generation in 0.6 and 6.9 ms, respectively. Their highly optimized algorithms, combined with lower computational costs, lead to these results. On the other hand, lattice-based post-quantum algorithms have much higher execution times: Dilithium2: 147.9 ms, Dilithium3: 289.6 ms, and Dilithium5: 320.8 ms. The signing times seem increased due to the internal operations of the lattice modules, which involve polynomial operations and modular arithmetic. Although slower than the NTRU-based methods, Dilithium2 offers promise as a compromise, delivering enough quantum security at a reasonable signing efficiency compared to higher-security levels.

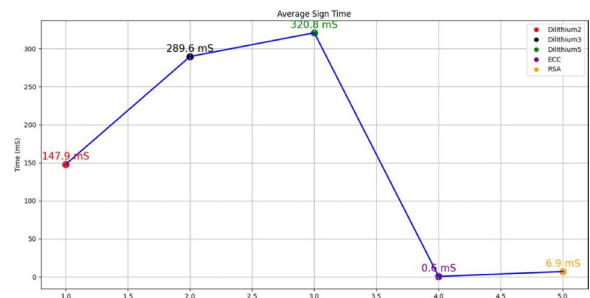
Signature verification is a common and time-sensitive operation in most cryptographic protocols, particularly in authentication-oriented applications. Again, as shown in Fig. 7, ECC and RSA outperform others, both achieving an



**Fig. 6** Average signature generation time for each algorithm



**Fig. 7** Average signature verification time for each algorithm



average of 0.5–0.6 ms, which suits high-frequency verification use cases. For lattice-based schemes, Dilithium2 comes out on top again in terms of efficiency, needing 50.7 ms on average, with Dilithium3 and Dilithium5 needing 79.4 and 123.1 ms, respectively: all slower than classical schemes, but for all Dilithium variants, verification times are within an acceptable range for many embedded applications. The reasonably linear scaling of verification time and security level also gives some freedom to designers to select a scheme that strikes a balance between performance and obstruction to quantum attack.

Taken as a whole, these results across the three performance dimensions (i.e., key generation, signing, and verification) support Dilithium2 as being the most efficient of the lattice-based candidates. It is extremely well-suited for practical deployment on constrained devices (e.g., Raspberry Pi), as it is a great trade-off between speed and post-quantum security. While ECC and RSA are faster than these schemes today, they do not stand up to quantum threats, which makes their long-term viability suspect. Correspondingly, the benchmarks demonstrate that lattice-based cryptography, and particularly the lower security levels, such as Dilithium2, are adequately on track for true analog embedded security applications with quantum safety requirements.

In addition to classical and quantum cryptanalytic attacks, modern threat models are also influenced by newly emerging threats like adversarial machine learning and poisoning attacks, as shown in Table 7. In distributed systems and federated learning settings, adversarial agents are capable of manipulating or inserting data to influence model behavior, often avoiding standard security perimeters. In these contexts, cryptographic mechanisms—post-quantum digital signatures, for example—underlie the security of model updates, identity verification for devices, and even the integrity of transmitted information. The implementation of Dilithium we have, with its highly secure unforgeability and tamper-resistance features, provides an essential layer of defense against evolving threats. Post-quantum signatures make sure that only authenticated and verified messages or models are accepted by IoT or edge devices, thus directly contributing to encouraging trust in AI-driven systems that are operating under adversarial conditions.

## 5.8 Discussion of limitations

Although this research illustrates the practicality of implementing lattice-based post-quantum signature schemes in embedded systems, we must consider limitations. Firstly, our evaluation was centered on the Dilithium family; other schemes such as Falcon or SPHINCS+ were not tested experimentally, despite their appealing features and trade-offs. Second, while we covered side-channel and fault injection risks, our implementation is not yet taking countermeasures such as masking, constant-time routines, or redundancies. Third, power consumption estimation was based on software profiling, instead of measuring directly using VCD/SAIF traces or hardware energy monitors, and thus, energy evaluation

**Table 7** Emerging security attacks and mitigation via post-quantum digital signatures

| Attack type                    | Description                                                          | Mitigation via PQ signatures (e.g., Dilithium)                               |
|--------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------|
| Model poisoning                | Injection of malicious data into training (e.g., federated learning) | Ensuring only authenticated and signed data/model updates are accepted       |
| Adversarial input manipulation | Manipulating input data to deceive ML models                         | Validates input integrity via signed data sources, preventing tampered input |
| Firmware/data injection        | Upload of unauthorized or tampered firmware or data                  | Signature checks block execution of unsigned or modified code/data           |
| Replay attacks                 | Reuse of previously valid messages to mislead systems                | Nonce-based signatures and freshness validation stop replayed messages       |
| Man-in-the-middle (MitM)       | Interception and tampering of messages between two endpoints         | Signature verification guarantees message authenticity and integrity         |
| Model theft or hijacking       | Illicit model copying or overwriting on edge devices                 | Signed model updates enable authentication and access control                |

accuracy was limited. Moreover, all our experiments were conducted on a single embedded platform (Raspberry Pi 4), and wider generalization to ultra-constrained microcontrollers or proprietary hardware (e.g., FPGAs) has yet to be investigated. We highlight these limitations and provide several opportunities for future work, such as improved security assessments, hardware-level analysis/profiling of feature extractors, and further algorithmic comparisons in conjunction with generalist models.

## 6 Conclusion and future work

In this article, we provided a practical implementation and benchmarking of lattice-based post-quantum digital signature schemes, Dilithium2, Dilithium3, and Dilithium5, also on the resource-constrained embedded platform, namely Raspberry Pi 4 Model B. Utilizing comprehensive performance analysis against classical cryptographic schemes (RSA and ECC), we proved that post-quantum cryptographic algorithms can be effectively deployed in low-power and embedded environments. While traditional schemes have relatively lower computational costs, especially in signature generation and verification, they lack strong post-quantum security guarantees, which is something that lattice-based schemes can offer. Of the entries tested, Dilithium2 stood out, providing a good balance between performance and security. Moreover, the graphical benchmarking interface we integrated makes our system user-friendly by allowing real-time testing and comparison of different cryptographic schemes.

These schemes were utilized in our previous exposure, the NIST signature competition; however, this is the first time they have been validated on a real-world embedded platform, as compared to prior works that are mainly theoretical frameworks or simulation environments. Although prior literature has focused on characterizing the underlying mathematical strength as well as refining specific protocol designs, our implementation showcases the pragmatics of applying different Dilithium variants in real-world scenarios. In particular, we demonstrate that Dilithium2 meets an excellent trade-off between security, latencies, and resource consumption and thus represents a reasonable candidate for post-quantum applications, e.g., IoT and embedded security. These results build on and mitigate previous work by anchoring post-quantum cryptography in observed phenomena, and our open-source toolset provides a reproducible foundation for ongoing research and development.

Directions for future research may include the following: You are also working on extending the benchmarking to other NIST finalist schemes like Falcon and SPHINCS+. Research on energy-aware optimizations of the signature generation on ultra-low-power microcontrollers (i.e., ESP32, STM32). Practical side-channel mitigation & analysis of retention to timing and fault injection attacks. Full stack IoT security testing by embedding lattice-based schemes into secure communication protocols (e.g., TLS, MQTT). With its experimental validation and open-access tooling, this work represents a significant advance toward composable quantum-secure cryptography in embedded systems in support of a worldwide shift to post-quantum security parameters.

To additionally provide the practicality and durability for post-quantum crypto-schemes, the next steps will research the following:

- Which includes benchmarking other post-quantum signature schemes like Falcon c and SPHINCS+ d on embedded platforms to evaluate implementation trade-offs in latency, memory footprint, and quantum resistance.
- Will also conduct hardware/software co-design with modern FPGA architecture (e.g., Xilinx Versal and AMD Zynq platforms) to accelerate specific lattice-based mathematical operations while maintaining the cryptographic properties of the FPGA, allowing for secure applications that can be implemented with low power.
- Working with real-time IoT protocols (MQTT, TLS, CoAP) to assess overall performance and security metrics in smart industrial, automotive, and healthcare scenarios.
- Application of techniques for protecting against physical-layer threats, including redundant computations, constant-time routines, and power analysis resistance methods
- Accurate power/energy profiling through VCD/SAIF simulation traces as well as real-time monitoring tools to quantize the energy efficiency for a given cryptographic workload.
- Evaluating the impact of different parameter sets, performance trade-offs, and architectures of post-quantum systems, providing recommendations on how to deal with the constraints of target devices.

**Author contributions** Mahmood A. Al-Shareeda: Conceptualization, Methodology, Software Development, Writing-Original Draft. Ahmed Abdul Hassan Ghadban: Software Implementation, Performance Benchmarking, Writing-Review & Editing. Ali Abdul Hadi Glass: Validation, Visualization, Code Optimization. Esam Mohamed Abdul Hadi: Data Analysis, Literature Review, Draft Review. Mohammed Amin Almaiah: Supervision, Project Administration, Final Review & Approval. All authors have read and approved the final manuscript.

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**Data availability** The data supporting the findings of this study are openly available in the GitHub repository at <https://github.com/ahmed-ghadban/Dilithium>. This includes the source code, implementation scripts, benchmarking tools, and performance logs used in this study.

**Clinical trial registration** Not applicable.

## Declarations

**Ethics approval and consent to participate** Not applicable.

**Consent for publication** Not applicable.

**Competing interests** The authors declare that they have no competing interests.

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## References

1. Barral D, Cardama FJ, Díaz-Camacho G, Failde D, Llovo IF, Mussa-Juane M, Vázquez-Pérez J, Villasuso J, Piñeiro C, Costas N, et al. Review of distributed quantum computing: from single QPU to high performance quantum computing. *Comput Sci Rev*. 2025;57: 100747.
2. Mousa RS, Shehab R. Applying risk analysis for determining threats and countermeasures in workstation domain. *J Cyber Secur Risk Audit*. 2025;2025(1):12–21.
3. Şafak I, Alagöz F, Anarim E. Post-quantum security measures for the internet of things. In: *Encyclopedia of information science and technology*. 6th ed. Hershey, PA: IGI Global; 2025. p. 1–44.
4. Näther C, Herzinger D, Gazdag S-L, Steghöfer J-P, Daum S, Loebenberger D. Migrating software systems towards post-quantum cryptography-a systematic literature review. *IEEE Access*. 2024;12:132107–13212.
5. Ootom S. Risk auditing for digital twins in cyber physical systems: a systematic review. *J Cyber Secur Risk Audit*. 2025;2025(1):22–35.
6. Sowa J, Hoang B, Yeluru A, Qie S, Nikolich A, Iyer R, Cao P. Post-quantum cryptography (PQC) network instrument: measuring PQC adoption rates and identifying migration pathways. *IEEE Int Conf Quantum Comput Eng*. 2024;1:1835–46.
7. Meng L, Fu Y, Zheng F, Wang M, Ma Z, Dong J, Lin J. HTM-PQC: hardening cryptography keys under the trend of post-quantum cryptography migration on industrial internet. *IEEE Trans Indust Informatics*. 2025;21(4):3504–14.
8. Almuqren AA. Cybersecurity threats, countermeasures and mitigation techniques on the IoT: future research directions. *J Cyber Secur Risk Audit*. 2025;1(1):1–11.
9. Kundalakesi M, Devi MR. Post-quantum cryptography methods. In: Raj P, Sundaravadivazhagan B, Ouaisa M, Kavitha V, Shantha Kumari K, editors. *Quantum computing and artificial intelligence: the industry use cases*. New York: Wiley; 2025. p. 77–97.
10. Al-Mekhlafi ZG, Al-Janabi HDK, Khalil A, Al-Shareeda MA, Mohammed BA, Alsadhan AA, Alayba AM, Saleh AMS, Al-Reshidi HA, Almekhlafi K. Lattice-based cryptography and fog computing based efficient anonymous authentication scheme for 5G-assisted vehicular communications. *IEEE Access*. 2024;12:71232–47.
11. Alshuaibi A, Almaayah M, Ali A. Machine learning for cybersecurity issues: a systematic review. *Secur Challenges*. 2025;1:2.
12. Köppl T, Zander R, Henkel L, Tcholtchev N. A parameter study for Ill and bkz with application to shortest vector problems. *arXiv preprint arXiv:2502.05160* (2025)
13. Liu F, Zheng Z, Gong Z, Tian K, Zhang Y, Hu Z, Li J, Xu Q. A survey on lattice-based digital signature. *Cybersecurity*. 2024;7(1):7.
14. Prajapat S, Gautam D, Kumar P, Jangirala S, Das AK, Park Y, Lorenz P. Secure lattice-based aggregate signature scheme for vehicular ad hoc networks. *IEEE Trans Vehicular Technol*. 2024;73(9):12370–84.
15. Al-Mekhlafi ZG, Al-Shareeda MA, Manickam S, Mohammed BA, Qtaish A. Lattice-based lightweight quantum resistant scheme in 5G-enabled vehicular networks. *Mathematics*. 2023;11(2):399.
16. Baklaga L. Quantum-resistant lattice-based cryptography: new conjectures on the learning with errors problem. *Scientific Practical Cyber Secur J*. 2024;81:50–6.
17. Sarkar P, Singha S. Classical reduction of gap SVP to LWE: a concrete security analysis. *Adv Math Commun*. 2023;17:484–99.
18. Anlet PJ, Jagadeeswari M. Parity based fault detection approach for the low power s-box and inverse s-box. 2012. <https://api.semanticscholar.org/CorpusID:18588105>.

19. Sarmadi SB, Kermani MM, Reyhani-Masoleh A. Efficient and concurrent reliable realization of the secure cryptographic SHA-3 algorithm. *IEEE Trans Comput Aided Des Integr Circuits Syst*. 2014;33:1105–9.
20. Canto AC, Kermani MM, Azarderakhsh R. Reliable code-based post-quantum cryptographic algorithms through fault detection on FPGA. 2023. In: 2023 IEEE Nordic Circuits and Systems Conference (NorCAS), p. 1–5.
21. Cintas-Canto A, Kermani MM, Azarderakhsh R. Error detection constructions for ITA finite field inversions over GF2 on FPGA using CRC and hamming codes. *IEEE Trans Reliab*. 2023;72:651–61.
22. Kermani MM, Azarderakhsh R, Mirakhorli M. Education and research integration of emerging multidisciplinary medical devices security. 2016. <https://api.semanticscholar.org/CorpusID:27392352>.
23. Türe ND, Cenk M. Efficient batch post-quantum signatures with crystals Dilithium. In: International Workshop on the Arithmetic of Finite Fields. Cham: Springer; 2024. p. 237–57.
24. Al-Shareeda MA, Gaber T, Alqarni MA, Alkinani MH, Almazroey AA, Almazroi AAA. Chebyshev polynomial based emergency conditions with authentication scheme for 5G-assisted vehicular fog computing. *IEEE Trans Dependable Secure Comput*. 2025. <https://doi.org/10.1109/TDSC.2025.355386>.
25. Dziechciarz D, Niemiec M. Efficiency analysis of NIST-standardized post-quantum cryptographic algorithms for digital signatures in various environments. *Electronics*. 2024;14(1):70.
26. Adapa VRK. Architecting quantum-resistant cybersecurity: a framework for transitioning to post-quantum cryptographic systems. *Int J Sci Res Archive*. 2025;14(1):737–46.
27. Zhang S, Li T, Sun Y, Wang B. A lattice-based anti-quantum privacy-preserving scheme for smart meter. *Int J Electr Power Energy Syst*. 2025;166:110502.
28. Wang Y, Li L, Zhou Y, Zhang H. A comprehensive review of MI-HFE and IPHFE cryptosystems: advances in internal perturbations for post-quantum security. *Axioms*. 2024;13(11):741.
29. Al Badawi A, Yeo SL, Yusof MFB. A generalized number-theoretic transform for efficient multiplication in lattice cryptography. *Contemp Math*. 2024;5(4):4200–22.
30. Yu Z. Research on cryptography based on quantum-resistant algorithms. 2024. In: 2024 International Conference on Electronics and Devices, Computational Science (ICEDCS), p. 149–154. IEEE
31. Doriguello JF, Giapitzakis G, Luongo A, Morolia A. On the practicality of quantum sieving algorithms for the shortest vector problem. 2024. arXiv preprint [arXiv:2410.13759](https://arxiv.org/abs/2410.13759).
32. Wu F, Zhou B, Song J, Xie L. Quantum-resistant blockchain and performance analysis. *J Supercomput*. 2025;81(3):498.
33. Wu F, Zhou B, Zhang X. Identity-based proxy signature with message recovery over NTRU lattice. *Entropy*. 2023;25(3):454.
34. Wu F, Zhou B, Jiang J, Lei T, Song J. Blockchain privacy protection based on post quantum threshold algorithm. *Comput Mater Continua*. 2023;76(1):957–73.
35. Jeudy C, Sanders O. Worst-case lattice sampler with truncated gadgets and applications. *Cryptology ePrint Archive*. 2024.
36. Häggblom C. Parameter selection in cryptography based on lattice isomorphism-cryptanalysis of the lattice isomorphism problem. 2024.
37. Jackson KA, Miller CA, Wang D. Evaluating the security of crystals-dilithium in the quantum random oracle model. In: Annual International Conference on the Theory and Applications of Cryptographic Techniques. Cham: Springer; 2024. p. 418–46.
38. Soni D, Basu K, Nabeel M, Manzano M, Karri R, Soni D, Basu K, Nabeel M, Aaraj N, Aaraj N, et al. Crystals-dilithium. Hardware architectures for post-quantum digital signature schemes. Cham: Springer; 2021. p. 13–30.
39. Ravi P, Chattopadhyay A, D'Anvers JP, Baksi A. Side-channel and fault-injection attacks over lattice-based post-quantum schemes (kyber, dilithium): survey and new results. *ACM Trans Embed Comput Syst*. 2024;23(2):1–54.
40. Land G, Sasdrich P, Güneysu T. A hard crystal-implementing dilithium on reconfigurable hardware. In: International Conference on Smart Card Research and Advanced Applications. Cham: Springer; 2021. p. 210–30.
41. Soni D, Basu K, Nabeel M, Aaraj N, Manzano M, Karri R, et al. Falcon. Hardware architectures for post-quantum digital signature schemes. Cham: Springer; 2021. p. 31–41.
42. Espitau T, Tibouchi M, Wallet A, Yu Y. Shorter hash-and-sign lattice-based signatures. In: Annual International Cryptology Conference. Cham: Springer; 2022. p. 245–75.
43. Fouque P-A, Hoffstein J, Kirchner P, Lyubashevsky V, Pornin T, Prest T, Ricosset T, Seiler G, Whyte W, Zhang Z, et al. Falcon: fast-fourier lattice-based compact signatures over NTRU. Submission to the NIST's post-quantum cryptography standardization process. 2018;36(5):1–75.
44. Yu H, Hui W. Certificateless ring signature from NTRU lattice for electronic voting. *J Inf Secur Appl*. 2023;75:103496.
45. Kirshanova E, May A, Nowakowski J. New NTRU records with improved lattice bases. In: International Conference on Post-Quantum Cryptography. Cham: Springer; 2023. p. 167–95.
46. Zheng X. Research on blockchain smart contract technology based on resistance to quantum computing attacks. *PLoS ONE*. 2024;19(5):0302325.
47. Sahu DR, Tiwari H, Tomar DS, Pateriya R. Quantum-resistant cryptography to prevent from phishing attack exploiting blockchain wallet. In: Sustainable security practices using blockchain, quantum and post-quantum technologies for real time applications. Cham: Springer; 2024. p. 171–91.
48. Monz T, Nigg D, Martinez EA, Brandl MF, Schindler P, Rines R, Wang SX, Chuang IL, Blatt R. Realization of a scalable Shor algorithm. *Science*. 2016;351(6277):1068–70.
49. Kumar M, Mondal B. Study on implementation of Shor's factorization algorithm on quantum computer. *SN Comput Sci*. 2024;5(4):413.
50. Shekhawat H, Gupta DS. Quantum-resistance blockchain-assisted certificateless data authentication and key exchange scheme for the smart grid metering infrastructure. *Pervasive Mob Comput*. 2024;100:101919.
51. Ravi PV, Chattopadhyay A, D'Anvers J-P, Baksi A. Side-channel and fault-injection attacks over lattice-based post-quantum schemes (kyber, dilithium): survey and new results. *ACM Trans Embedded Comput Syst*. 2023;23:1–54.